

# Portfolio Governance Bot

*A dual-entry (Slack + HTML) chatbot answering portfolio/program/project/Epics/Sub-tasks  
-status questions against a live sample Google Sheet*

n8n

GPT-4o-mini

Slack API

Google Sheets

# Problem & Context

1

## The ask

Build a Portfolio Governance Digest: multiple flows, each processing a source, synthesized into one digest delivered to a channel.

2

## The gap

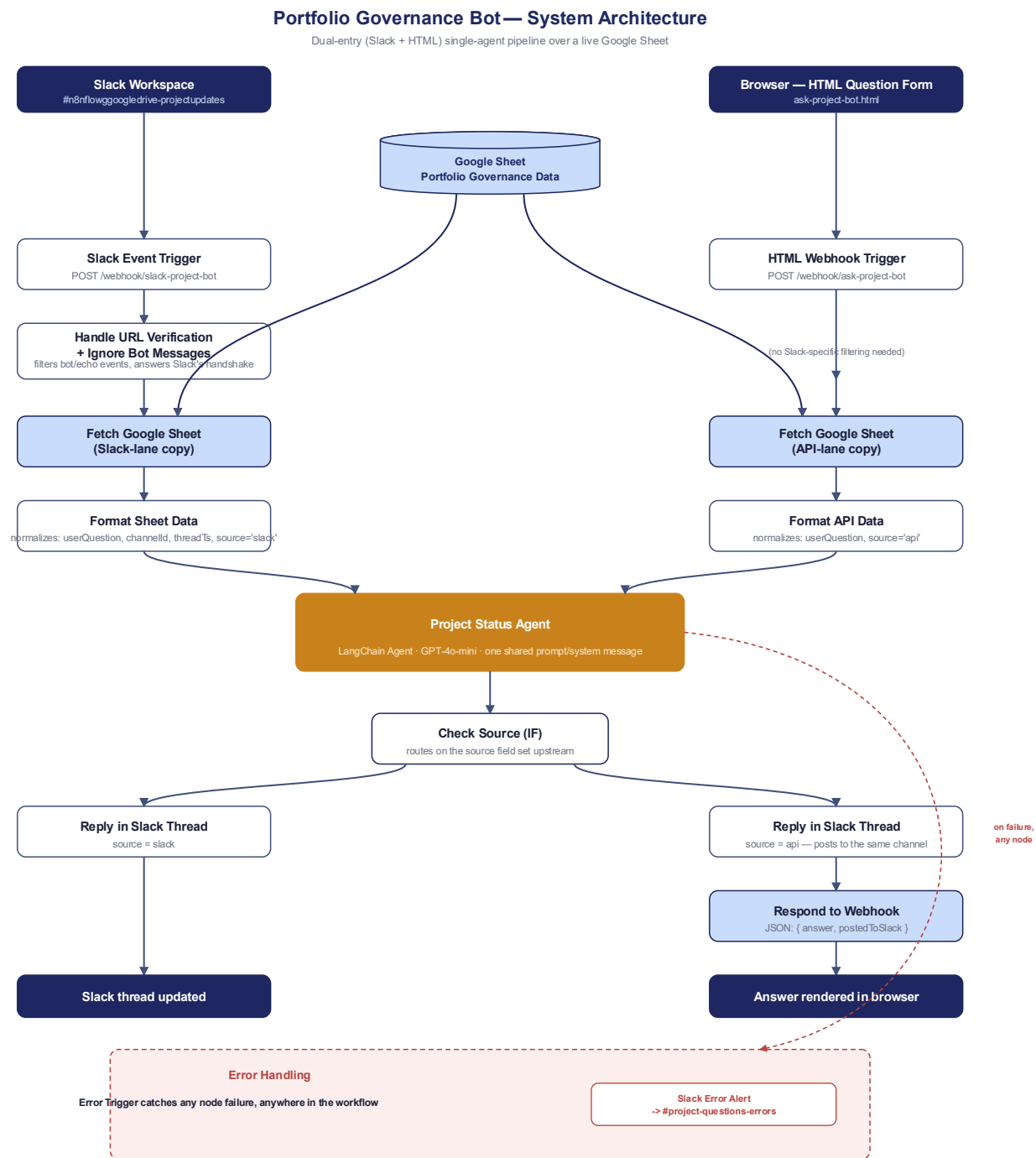
Portfolio managers re-read the same tracker repeatedly for the same questions: what's at risk, what's slipped, what needs governance attention.

3

## This project

A deliberately scoped single-agent step first; proving the trigger, filtering, fetch, reasoning, and error-handling plumbing a multi-agent system still needs.

# System Flow



# Key Decisions & Trade-offs

1

## **n8n over a custom backend**

Faster to iterate visually — traded for UI-driven bugs (stray characters, mislabeled nodes) being harder to catch than compiler errors.

2

## **Two parallel ingestion lanes, not one merged lane**

Slack and HTML paths each get their own Fetch/Format nodes, converging only at the Agent — avoids cross-branch reference errors.

3

## **One shared Agent for both surfaces**

Both lanes normalize to one shape so a single gpt-4o-mini, Agent serves both

4

## **gpt-4o-mini, not a larger model**

Bounded, structured Q&A over a spreadsheet doesn't need a frontier model; cost and latency mattered more.

5

## **No auth on either webhook — stated, not hidden**

An explicit trade-off for a scoped demo, flagged plainly as the first thing to fix before production traffic.

## Sample Queries

*Pls share the high-risk projects and state why? Also, share overall assessment of projects for this portfolio.*

*Which projects are running in March? Any severity/dependencies?*

*Show which projects crossed their baseline due to bugs or specific issues.. mention their start and end dates. Mention the risks associated, bugs to fix, issues to fix and finally anything specific to review for Governance against these projects..*

# Outcome

## What held up

- ✓ Dual-entry, single-agent design worked once wired correctly — one prompt, one model, two front doors.
- ✓ Formula-driven rollups (RAG status, days slipped, % complete) kept governance answers accurate with no hand-maintained summaries.
- ✓ Workflow-level error handling (one Error Trigger, one alert channel) was enough to catch failures at this scale.

## What I'd do next

- Fix the Respond-setting/duplicate-node class of bug earlier by auditing node types, not node names.
- Add authentication to both webhooks before any real traffic touches them.
- Move JSON responses to object expressions everywhere, not string templates, to avoid the multiline/quote failure mode.